

Current Score: 0/26

Question	1	2	3	4	5	6	7	8	9	Total
Points	0/3	0/2	0/2	0/8	0/2	0/2	0/1	0/1	0/5	0/26

Description

This test is the WebAssign, Part A, of the Semester Test 2. It carries 26 points. Part A and Part B contribute 50:50 to Semester Test 2. Part B will be released at 14:45 in ClickUP, from the "Semester and Online Tests Folder" section for Part B. The time allocation for Part B is 120 minutes. ie. from 14:45 to 16:45.

Instructions

*Answer all questions. *Give your final answers to the correct number of significant figures. *Your final answers for large and small numbers need to be written in scientific notation. (In WebAssign the format for example for representing 2.87×10^{-5} is 2.87×10^{-5}). *Insert a space between your numerical answer and the units if units are needed. *Also use the PhysPad/MathsPad for mathematical layouts, such as roots, fractions, superscripts and subscripts etc. *In symbolic format, for example, $2mg$ is $2 \cdot m \cdot g$ where the $\cdot$ is the dot / multiply operator. *Use a laptop computer instead of a smartphone, for, smartphones do not work with the PhysPad/MathsPad. *This is a timed assessment. *You have 60 minutes (1 hour) to complete Part A. ie. deadline is 14h30. *Just before your final submission, print out your now filled-in WebAssign Part A test (there is a "print" tab at the top of the test, it prints to a PDF file - you don't need a physical printer!). *Submit it into the relevant part in the "Semester and Online Tests Folder" on ClickUP. *Do this submission immediately after submitting the WebAssign Part A test. This is what the 15 min gap is for between the end of Part A and the beginning of Part B. *NO late submissions allowed! *You have 2 submission opportunities in WebAssign - but these must be completed within the allotted time for this activity.

1. 0/3 points

A plane is flying to a city **776** km directly north of its initial location. The plane maintains a speed of **188** km/h relative to the air during its flight.

- (a) If the plane flies through a constant headwind blowing south at **55.5** km/h, how much time (in h) will it take to reach the city?

  **5.86** h

- (b) If instead the plane flies through a constant tailwind blowing at **55.5** km/h, how much time (in h) will it take to reach the city?

  **3.19** h

- (c) If instead the plane flies through a constant crosswind blowing east at **55.5** km/h, how much time (in h) will it take to reach the city?

  **4.32** h

Solution or Explanation

Note: We are displaying rounded intermediate values for practical purposes. However, the calculations are made using the unrounded values.

The airplane (AP) travels through the air (W) that can move relative to the ground (G). The airplane is to make a displacement of **776** km north. Treat north as positive y and west as positive x.

- (a) The wind (W) is blowing at **55.5** km/h, south. The northern component of the airplane's velocity relative to the ground is

$$(v_{AP, G})_y = (v_{AP, W})_y + (v_{W, G})_y = 188 \text{ km/h} - 55.5 \text{ km/h} = 133 \text{ km/h}.$$

We can find the time interval the airplane takes to travel **776** km north.

$$\Delta y = (v_{AP, G})_y \Delta t \rightarrow \Delta t = \frac{\Delta y}{(v_{AP, G})_y} = \frac{776 \text{ km}}{133 \text{ km/h}} = 5.86 \text{ h}.$$

- (b) The wind (W) is blowing at **55.5** km/h, north. The northern component of the airplane's velocity relative to the ground is

$$(v_{AP, G})_y = (v_{AP, W})_y + (v_{W, G})_y = 188 \text{ km/h} + 55.5 \text{ km/h} = 244 \text{ km/h}.$$

We can find the time interval the airplane takes to travel **776** km north.

$$\Delta t = \frac{\Delta y}{(v_{AP, G})_y} = \frac{776 \text{ km}}{244 \text{ km/h}} = 3.19 \text{ h}.$$

- (c) Now, the wind (W) is blowing at **55.5** km/h, east. The airplane must travel directly north to reach its destination, so it must head somewhat west and north so that the east component of the wind's velocity is canceled by the airplane's west component of velocity. If the airplane heads at an angle θ measured west of north, then

$$\begin{aligned} (v_{AP, G})_x &= (v_{AP, W})_x + (v_{W, G})_x \\ &= (188 \text{ km/h}) \sin(\theta) + (-55.5 \text{ km/h}) = 0 \\ \sin(\theta) &= \frac{55.5 \text{ km/h}}{188 \text{ km/h}} \rightarrow \theta = 17.2^\circ. \end{aligned}$$

The northern component of the airplane's velocity relative to the ground is

$$(v_{AP, G})_y = (v_{AP, W})_y + (v_{W, G})_y = (188 \text{ km/h}) \cos(17.2^\circ) + 0 = 180 \text{ km/h}.$$

We can find the time interval the airplane takes to travel **776** km north.

$$\Delta t = \frac{\Delta y}{(v_{AP, G})_y} = \frac{776 \text{ km}}{180 \text{ km/h}} = 4.32 \text{ h}$$

Need Help?

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2. 0/2 points

A 14.5 g bullet is moving to the right with speed 250 m/s when it hits a target and travels an additional 23.4 cm into the target. What are the magnitude (in N) and direction of the stopping force acting on the bullet? Assume the stopping force is constant.

magnitude   1940 N
 direction ---Select---   to the left

Solution or Explanation

Note: We are displaying rounded intermediate values for practical purposes. However, the calculations are made using the unrounded values.

Use $v_f^2 = v_i^2 + 2a(x_f - x_i)$ with $v_f = 0$ and $\Delta x = x_f - x_i = 0.234$ m to find a .

$$0 = v_i^2 + 2a\Delta x \rightarrow a = -\frac{v_i^2}{2\Delta x} = -\frac{(250 \text{ m/s})^2}{2(0.234 \text{ m})} = -1.34 \times 10^5 \text{ m/s}^2$$

In the target, the only force acting on the bullet is the kinetic stopping force. Apply Newton's second law to the bullet.

$$\sum F_x = ma_x \rightarrow f_k = ma$$

Convert the bullet's mass to kg and insert values.

$$f_k = (0.0145 \text{ kg})(-1.34 \times 10^5 \text{ m/s}^2) = -1,940 \text{ N}$$

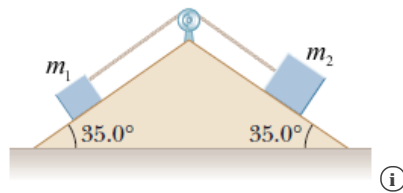
The magnitude of the stopping force is 1,940 N, and the direction is opposite the direction of motion, as we see from the minus sign, which means to the left.

Need Help?

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3. 0/2 points

A cord passing over a pulley connects two masses, as shown, where $m_1 = 3.90$ kg and $m_2 = 7.00$ kg. The system accelerates with a magnitude of 1.12 m/s². The coefficient of kinetic friction between the masses and the incline is the same for both masses. Assume the pulley is frictionless, and the cord is massless. (Due to the nature of this problem, do not use rounded intermediate values in your calculations—including answers submitted in WebAssign.)



(a) What is the coefficient of kinetic friction?

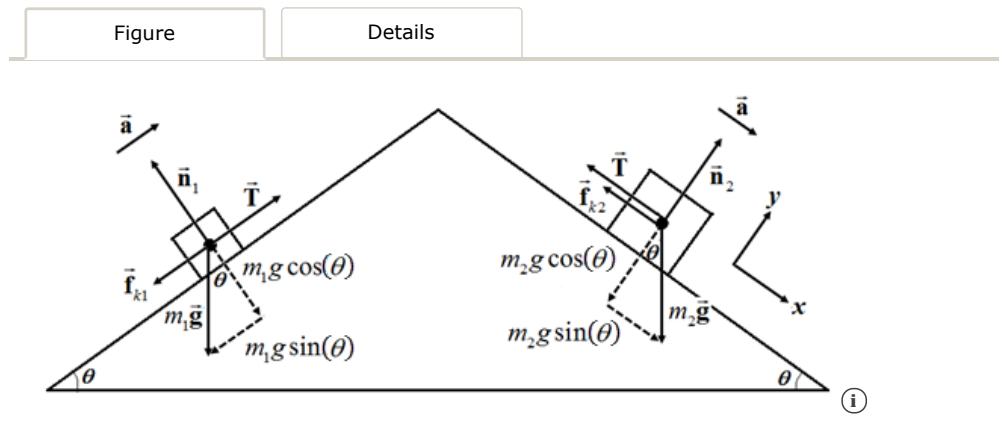
✗ 0.0596

(b) What is the tension (in N) in the cord?

✗ 28.2 N

Solution or Explanation

(a) Draw free-body diagrams for both blocks. Note the choice of axes for each block. We can assume the acceleration of the system is to the right, because $m_2 > m_1$.



Write Newton's second law in the x-direction for each mass.

For m_1 ,

$$\sum F_x = m_1 a = T - m_1 g \sin(35^\circ) - f_{k1} = m_1 a.$$

For m_2 ,

$$\sum F_x = m_2 a = -T - f_{k2} + m_2 g \sin(35^\circ) = m_2 a.$$

Solving Newton's second law in the y-direction gives the following.

For m_1 ,

$$\sum F_y = 0 \rightarrow n_1 = m_1 g \cos(\theta).$$

For m_2 ,

$$\sum F_y = 0 \rightarrow n_2 = m_2 g \cos(\theta).$$

Note that $f_k = \mu_k n$, so the second-law equations in the x-direction are

$$\begin{aligned} T - m_1 g \sin(35^\circ) - \mu_k m_1 g \cos(35^\circ) &= m_1 a \\ -T - \mu_k m_2 g \cos(35^\circ) + m_2 g \sin(35^\circ) &= m_2 a. \end{aligned}$$

Solve this system of equations.

$$\begin{array}{r}
 T - m_1 g \sin(35^\circ) - \mu_k m_1 g \cos(35^\circ) = m_1 a \\
 -T - \mu_k m_2 g \cos(35^\circ) + m_2 g \sin(35^\circ) = m_2 a \\
 \hline
 (m_2 - m_1) g \sin(35^\circ) - (m_1 + m_2) \mu_k g \cos(35^\circ) = (m_1 + m_2) a
 \end{array}$$

Solving for μ_k , we get

$$\mu_k = \frac{(m_2 - m_1) g \sin(35^\circ) - (m_1 + m_2) a}{(m_1 + m_2) g \cos(35^\circ)}.$$

Substituting values,

$$\mu_k = \frac{((7.00 \text{ kg}) - (3.90 \text{ kg}))(9.80 \text{ m/s}^2) \sin(35^\circ) - ((3.90 \text{ kg}) + (7.00 \text{ kg}))(1.12 \text{ m/s}^2)}{((3.90 \text{ kg}) + (7.00 \text{ kg}))(9.80 \text{ m/s}^2) \cos(35^\circ)} = 0.0596.$$

(b) To calculate the tension, use either second-law equation and solve for T .

$$T - m_1 g \sin(35^\circ) - \mu_k m_1 g \cos(35^\circ) = m_1 a \rightarrow T = m_1 (g \sin(35^\circ) + \mu_k g \cos(35^\circ) + a)$$

Substitute your expression for μ_k .

$$\begin{aligned}
 T &= m_1 \left(g \sin(35^\circ) + \left(\frac{(m_2 - m_1) g \sin(35^\circ) - (m_1 + m_2) a}{(m_1 + m_2) g \cos(35^\circ)} \right) g \cos(35^\circ) + a \right) \\
 &= m_1 \left(g \sin(35^\circ) + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g \sin(35^\circ) - a + a \right) \\
 &= m_1 g \sin(35^\circ) \left(1 + \frac{m_2 - m_1}{m_1 + m_2} \right) \\
 &= m_1 g \sin(35^\circ) \left(\frac{2m_2}{m_1 + m_2} \right)
 \end{aligned}$$

Plugging in values, we get

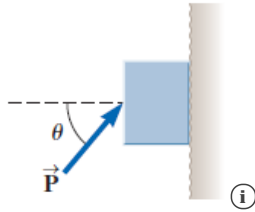
$$T = (3.90 \text{ kg})(9.80 \text{ m/s}^2) \sin(35^\circ) \left(\frac{2(7.00 \text{ kg})}{(3.90 \text{ kg}) + (7.00 \text{ kg})} \right) = 28.2 \text{ N}.$$

Need Help?

Read It

4. 0/8 points

A block of mass **3.20** kg is pushed up against a wall by a force $\vec{P}$ that makes an angle of $\theta = 50.0^\circ$ angle with the horizontal as shown below. The coefficient of static friction between the block and the wall is **0.260**.



(a) Determine the possible values for the magnitude of $\vec{\mathbf{P}}$ that allow the block to remain stationary. (If there is no maximum, enter NONE in that answer blank.)

$|\vec{\mathbf{P}}_{\text{max}}| =$   52.4 N

$|\mathbf{p}_{\min}| =$   33.6 N

(b) What happens if $|\vec{\mathbf{P}}|$ has a larger value than $|\vec{\mathbf{P}}|_{\text{max}}$?

- ☐  The block slides up the wall.
- ☐ The block does not slide along the wall.
- ☐ The block slides down the wall.

What happens if $|\mathbf{P}|$ has a smaller value than $|\mathbf{P}_{\min}|$?

- ☐ The block slides up the wall.
- ☐ The block does not slide along the wall.
- ☒ The block slides down the wall.

(c) Repeat parts (a) and (b) assuming the force makes an angle of $\theta = 12.4^\circ$ with the horizontal. Determine the possible values for the magnitude of $\vec{\mathbf{P}}$ that allow the block to remain stationary. (If there is no maximum, enter NONE in that answer blank.)

$|\vec{\mathbf{p}}_{\text{max}}| =$ NONE N

$|\mathbf{p}_{\min}| = \boxed{} \times \boxed{66.9} \text{ N}$

What happens if $|\vec{\mathbf{P}}|$ has a larger value than $|\vec{\mathbf{P}}_{\max}|$?

- ☐ The block slides up the wall.
- ☒  The block does not slide along the wall.
- ☐ The block slides down the wall.

What happens if $|\mathbf{P}|$ has a smaller value than $|\mathbf{P}_{\min}|$?

- ☐ The block slides up the wall.
- ☐ The block does not slide along the wall.
- ☒  The block slides down the wall.

Solution or Explanation

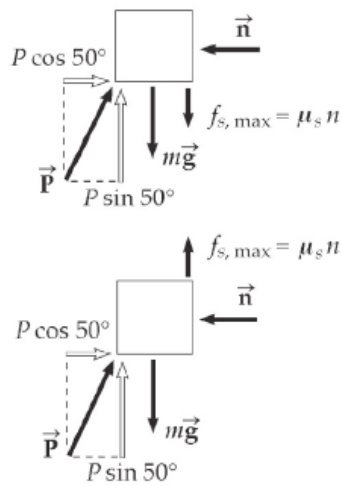
Note: We are displaying rounded intermediate values for practical purposes. However, the calculations are made using the unrounded values.

Consider the figure below.

Figure

Upward Motion

Downward Motion



(a) To find the maximum possible value of P , imagine impending upward motion as case 1. Setting $\sum F_x = 0$:

$$P \cos 50.0^\circ - n = 0$$

with $f_{s, \max} = \mu_s n$:

$$\begin{aligned} f_{s, \max} &= \mu_s P \cos 50.0^\circ \\ &= 0.260(0.643)P = 0.167P. \end{aligned}$$

Setting $\sum F_y = 0$:

$$\begin{aligned} P \sin(50.0^\circ) - 0.167P - (3.20 \text{ kg})(9.80 \text{ m/s}^2) &= 0 \\ P_{\max} &= 52.4 \text{ N}. \end{aligned}$$

To find the minimum possible value of P , consider impending downward motion. As in case 1,

$$f_{s, \max} = 0.167P.$$

Setting $\sum F_y = 0$:

$$\begin{aligned} P \sin(50.0^\circ) + 0.167P - (3.20 \text{ kg})(9.80 \text{ m/s}^2) &= 0 \\ P_{\min} &= 33.6 \text{ N}. \end{aligned}$$

(b) If $P > 52.4 \text{ N}$, the block slides up the wall. If $P < 33.6 \text{ N}$, the block slides down the wall.

(c) We repeat the calculation as in part (a) with the new angle. Consider impending upward motion as case 1.

Setting $\sum F_x = 0$:

$$P \cos(12.4^\circ) - n = 0$$

with $f_{s, \max} = \mu_s n$:

$$\begin{aligned} f_{s, \max} &= \mu_s P \cos(12.4^\circ) \\ &= 0.260(0.977)P = 0.254P. \end{aligned}$$

Setting $\sum F_y = 0$:

$$\begin{aligned} P \sin(12.4^\circ) - 0.254P - (3.20 \text{ kg})(9.80 \text{ m/s}^2) &= 0 \\ P_{\max} &= -800 \text{ N}. \end{aligned}$$

The push cannot really be negative. However large or small it is, it cannot produce upward motion. To find the minimum possible value of P , consider impending downward motion. As in case 1,

$$f_{s, \max} = 0.254P.$$

Setting $\sum F_y = 0$:

$$\begin{aligned} P \sin(12.4^\circ) + 0.254P - (3.20 \text{ kg})(9.80 \text{ m/s}^2) &= 0 \\ P_{\min} &= 66.9 \text{ N}. \end{aligned}$$

$P \geq 66.9 \text{ N}$. The block cannot slide up the wall. If $P < 66.9 \text{ N}$, the block slides down the wall.

Need Help?

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5. 0/2 points

- (a) A stuntman with a mass of **82.0** kg swings across a pond from a rope that is **10.5** m. At the bottom of the swing the stuntman's speed is **7.95** m/s. The rope's breaking strength is 1,000 N. Will the stuntman make it across the pond without falling in?

☐ Yes
☒ No

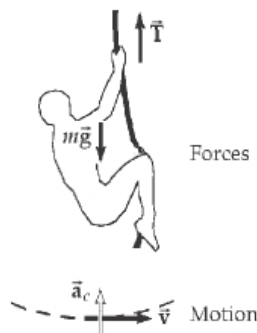


- (b) **What If?** What is the maximum speed (in m/s) that the stuntman can have at the bottom of the swing on this vine to safely swing across the river?

5.01 m/s

Solution or Explanation

- (a) Consider the system to be the stuntman when he is at the very bottom of the swing. At this point, there are two force acting on him: the gravitational force downward (magnitude mg) and the tension force due to the rope upward (magnitude T). There are no other forces acting on the system.



(i)

The stuntman's acceleration is upward at this instant (and therefore is the centripetal acceleration a_c), and his velocity v , as shown above, is to the right, perpendicular to the acceleration.

If the positive y -direction is upward, we can then apply Newton's second law on the y -axis and solve for the tension force.

$$\begin{aligned}\sum F_y &= ma_y \\ T - mg &= ma_c \\ T - mg &= \frac{mv^2}{r} \\ T &= m\left(g + \frac{v^2}{r}\right) \\ T &= (82.0 \text{ kg})\left(9.80 \text{ m/s}^2 + \frac{(7.95 \text{ m/s})^2}{10.5 \text{ m}}\right) = 1,300 \text{ N}\end{aligned}$$

The tension force in the rope is greater than its breaking strength of 1,000 N. Therefore the rope will break, and the stuntman will not make it across.

- (b) Solution to **What If?**

In part (a), we found:

$$T - mg = \frac{mv^2}{r}$$

We can solve this for v , where now T is the maximum possible tension of 1,000 N.

$$\begin{aligned}v &= \sqrt{r\left(\frac{T}{m} - g\right)} \\ v &= \sqrt{(10.5 \text{ m})\left(\frac{1,000 \text{ N}}{82.0 \text{ kg}} - 9.80 \text{ m/s}^2\right)} = 5.01 \text{ m/s}\end{aligned}$$

Need Help?

Read It

6. 0/2 points

SerPSE10 6.4.OP.015. [4004880]

- (a) A sphere made of **plastic** has a density of **1.06** g/cm³ and a radius of **8.00** cm. It falls through air of density 1.20 kg/m³ and has a drag coefficient of 0.500. What is its terminal speed (in m/s)?

   **60.8** m/s

- (b) From what height (in m) would the sphere have to be dropped to reach this speed if it fell without air resistance?

   **188** m

Solution or Explanation

- (a) At terminal speed, the object moves at a constant speed, so the acceleration is zero. The air resistance force is upward and the weight force is downward. From Newton's second law,

$$\sum F_y = 0$$

$$R - mg = 0$$

Plugging in the formula for the air resistance force R ,

$$\frac{1}{2} D \rho_{\text{air}} A v^2 - mg = 0.$$

Solving for the speed v ,

$$v = \sqrt{\frac{2mg}{D \rho_{\text{air}} A}}$$

where ρ_{air} is the density of air, A is the cross sectional area of the falling sphere, and D is the drag coefficient. If r is the radius of the sphere,

$$v = \sqrt{\frac{2mg}{D \rho_{\text{air}} \pi r^2}}.$$

The mass of the sphere depends on its density ρ and volume.

$$v = \sqrt{\frac{2\rho \left(\frac{4}{3}\pi r^3\right) g}{D \rho_{\text{air}} \pi r^2}}$$

$$v = \sqrt{\frac{8\rho r g}{3D \rho_{\text{air}}}}$$

Plugging in values and converting units,

$$v = \sqrt{\frac{8(1.06 \text{ g/cm}^3) \left(\frac{1 \text{ kg}}{1000 \text{ g}}\right) \left(\frac{100 \text{ cm}}{1 \text{ m}}\right)^3 (8.00 \times 10^{-2} \text{ m})(9.80 \text{ m/s}^2)}{3(0.500)(1.20 \text{ kg/m}^3)}}$$

$$v = \mathbf{60.8 \text{ m/s}}$$

- (b) We can use the constant acceleration equation that relates initial speed, final speed, acceleration, and distance in the y -direction. Note the sphere starts from rest.

$$v_{yf}^2 = v_{yi}^2 + 2a_y(y_f - y_i)$$

$$v_{yf}^2 = -2g(-y_i)$$

$$y_i = \frac{v_{yf}^2}{2g}$$

$$y_i = \frac{(60.8 \text{ m/s})^2}{2(9.80 \text{ m/s}^2)} = \mathbf{188 \text{ m.}}$$

Need Help?

Read It

7. 0/1 points

SerPSE10 7.4.P.012. [4037860]

Express the units of the force constant of a spring in SI fundamental units. (Use m for meter, kg for kilogram, and s for second in your equation.)

$[k] =$ $\times$ $k \frac{g}{s^2}$

Need Help?

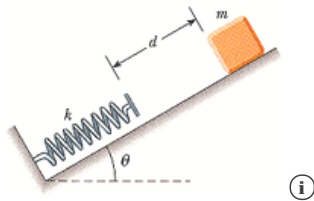
Read It

8. 0/1 points

SerPSE10 8.XP.005. [4212710]

An object of mass m starts from rest and slides a distance d down a frictionless incline of angle θ . While sliding, it contacts an unstressed spring of negligible mass as shown in the figure. The object slides an additional distance x as it is brought momentarily to rest by compression of the spring (of force constant k). Find the initial separation d between object and spring. (Use any variable or symbol stated above along with the following as necessary: g .)

$d =$ $\times$ $k \cdot \frac{x^2}{2 \cdot m \cdot g \cdot \sin(\theta)} - x$



(i)

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9. 0/5 points

SerPSE10 8.XP.008. [4041872]

A single constant force $\mathbf{F} = (3 \mathbf{i} + 4 \mathbf{j})$ N acts on a 3.95 kg particle.

(a) Calculate the work done by this force if the particle moves from the origin to the point having the vector position $\mathbf{r} = (2 \mathbf{i} - 3 \mathbf{j})$ m.

 $\times$ J

-6

Does this result depend on the path?

☐ Yes

☐ No

Explain.

Key: The result does Not depend on the path since the force is conservative.

(b) What is the speed of the particle at $\mathbf{r}$ if its speed at the origin is 4 m/s?

 $\times$

3.6

m/s

(c) What is the change in the potential energy?

 $\times$

6

J

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Assignment Details

Name (AID): FSK116 Part A of Semester Test 2 (18940258)

Submissions Allowed: 2

Category: Test

Code:

Locked: Yes

Feedback Settings

Before due date

After due date

Author: **Legodi, MJ (Matshisa)** (matshisa.legodi@up.ac.za)

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